



UNIVERSITY OF GENEVA

Faculty of Science

Exercise #3

Introduction to Computational Finance
14X030

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1 Seasonality

The file `eur_usd_20120101_20120301.txt` (download it on Moodle) contains data on the EUR / USD exchange rate for the period of January 1st to March 1st 2012, with the following data structure:

```
timestamp    bid    ask
```

Where *timestamp* denotes the time (in seconds) measured starting from 01 Jan 1970 00:00:00.

Considering bins of 15 minutes, plot the two histograms of the daily and weekly distributions of ticks. Comment on the obtained histograms.

```
import matplotlib.pyplot as plt
import matplotlib.dates as mdates
import numpy as np
import datetime

data = np.loadtxt('eur_usd_20120101_20120301.txt')
dates = np.array([datetime.datetime.fromtimestamp(x) for x in data[:, 0]])

bins = np.arange(0, 24*60 * 60, 15 * 60)

print(len(bins))
# day_1 = data[data[:, 0] < data[0, 0] + 24*60*60, 0]
# week_1 = data[data[:, 0] < data[0, 0] + 7*24*60*60, 0]

# def selection(data: np.ndarray, day_number: int = 1, week: bool = False):
#     mask = data[:, 0] // (24*3600)
#     mask -= mask[0]
#     if week:
#         result = data[day_number <= mask < day_number + 7, 0]
#     else:
#         result = data[mask == day_number, 0]
#     return np.array([datetime.datetime.fromtimestamp(x) for x in result])
#     # return result % (24 * 3600)

def selection(dates: list, day_number: int = 1, week: bool = False) -> list:
    first_day = dates[0].date() + datetime.timedelta(days=1)
    if week:
        return [d for d in dates if first_day <= d.date() < first_day +
datetime.timedelta(days=7)]
    return [d for d in dates if d.date() == first_day]

plt.figure()
plt.hist(selection(dates, day_number=1), bins=len(bins))
plt.gca().xaxis.set_major_locator(mdates.HourLocator())
plt.gca().xaxis.set_major_formatter(mdates.DateFormatter('%H:%M'))
plt.legend()
plt.title("Test")
```



```
plt.gcf().autofmt_xdate()
plt.show()
```

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```
[stderr]
<block 1>:35: UserWarning: No artists with labels found to put in legend. Note
that artists whose label start with an underscore are ignored when legend() is
called with no argument.
```

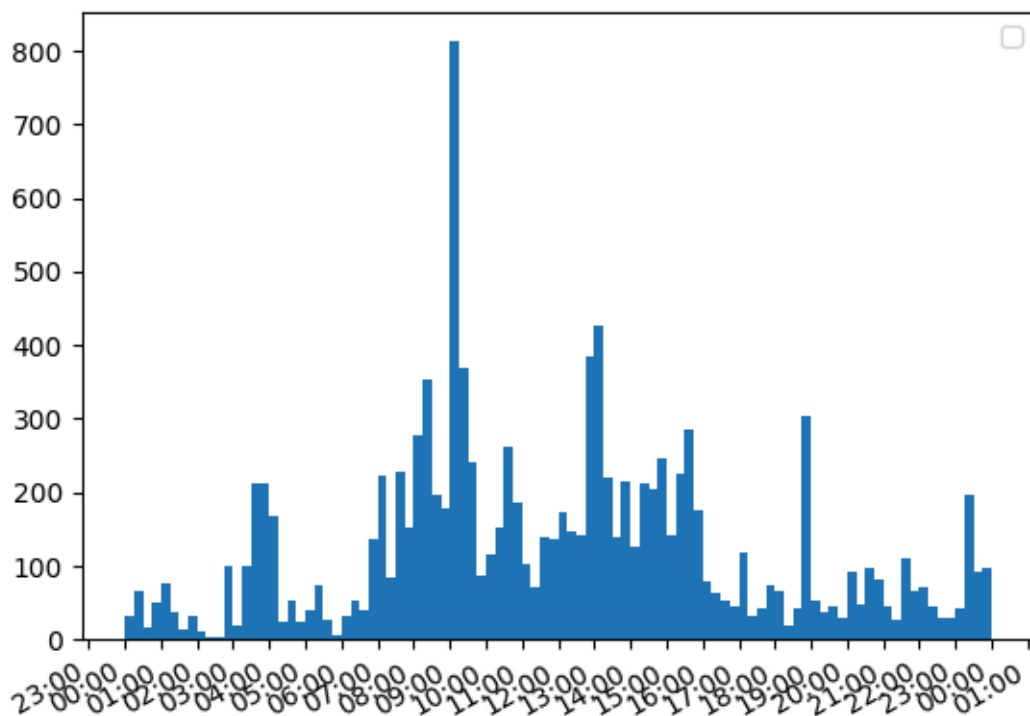


Figure 1: Test

2 AR(1) process

For different values of φ_1 , simulate and plot an AR(1) process: $X_t = \varphi_1 X_{t-1} + \varepsilon_t$ with $X_0 = 0$ and $\varepsilon_t \sim \mathcal{N}(0, 1)$.

In particular, how do you obtain:

- A stationary process?
- A mean-reverting process?
- A trendy process?
- An exploding process?



3 Mean-Reversion in Finance: The discrete Vasicek model

In the previous exercise, you explored the theoretical properties of the AR(1) process. In quantitative finance, while stock log-prices p_t (where $p_t = \ln(S_t)$ and S_t is the stock price) are often modeled as random walks (a non-stationary AR(1) process with $\varphi_1 = 1$), interest rates are structurally different: they typically cannot grow indefinitely but instead fluctuate around a long-term macroeconomic equilibrium rate.

One of the foundational models capturing this phenomenon is the Vasicek model for the short-term interest rate. In its discrete-time formulation, the interest rate r_t evolves according to:

$$r_t - r_{t-1} = \kappa (\theta - r_{t-1}) + \sigma \varepsilon_t \quad (1)$$

where $\varepsilon_t \sim \mathcal{N}(0, 1)$, κ represents the speed of mean reversion, θ is the long-term average level, and σ is the volatility of the rate.

Answer the following questions:

- 1. Autoregressive Representation:** Show that the discrete Vasicek model can be algebraically rewritten as a classical AR(1) process with drift, of the form $r_t = c + \varphi_1 r_{t-1} + \sigma \varepsilon_t$. Explicitly identify the constants c and φ_1 in terms of the initial Vasicek parameters κ and θ .
- 2. Mean-Reversion Condition:** Based on your findings from the previous AR(1) exercise (specifically, the mean-reverting process case), what strict mathematical boundaries must the speed of mean reversion κ satisfy for the interest rate process r_t to be weakly stationary? What are the financial reality and the dynamics of r_t if $\kappa = 0$?
- 3. Long-term Equilibrium:** Assuming the stationarity condition holds, rigorously compute the unconditional expectation $E[r_t]$ and the unconditional variance $\text{Var}(r_t)$ as $t \rightarrow \infty$. Discuss how the parameter θ dictates the long-term properties of the interest rates model.
- 4. Simulation and Visualization:** Simulate a single path of the discrete Vasicek model over $T = 300$ time steps using the following parameters: $r_0 = 0.05$, $\kappa = 0.1$, $\theta = 0.05$, and $\sigma = 0.01$. Plot the simulated interest rate r_t against time t , and overlay a horizontal line representing the long-term equilibrium rate θ . Comment on the observed dynamics, contrasting them with a classic random walk.

4 Vocabulary

1. Describe what are bonds and stocks and the difference between them.
2. What is an option? A future?



3. What is an index? What are the typical methods to weight the items of an index? For instance, what method is used by Nasdaq and S&P 500?